

46-921, Homework 3 Solutions

1. Note that we will use the parametrization of the Gamma distribution that gives the following pdf:

$$f(x; \alpha, \beta) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}, x > 0 \quad (1)$$

We have that the following facts are known for $X \sim \text{Gamma}(\alpha, \beta)$ distribution:

- $\mathbb{E}[X] = \frac{\alpha}{\beta}$;
- $\mathbb{V}[X] = \frac{\alpha}{\beta^2}$.

Hence we have that:

$$\mathbb{E}[X^2] = \mathbb{V}[X] + (\mathbb{E}[X])^2 = \frac{\alpha}{\beta^2} + \frac{\alpha^2}{\beta^2} = \frac{\alpha}{\beta^2}(1 + \alpha) \quad (2)$$

So if we set up the equations considering the first two moments:

$$\begin{cases} \frac{\alpha}{\beta} = \frac{1}{n} \sum_{i=1}^n X_i \\ \frac{\alpha}{\beta^2}(1 + \alpha) = \frac{1}{n} \sum_{i=1}^n X_i^2 \end{cases} \quad (3)$$

$$\Rightarrow \begin{cases} \alpha = \bar{X}\beta \\ \frac{\bar{X}}{\beta} = \frac{1}{n} \sum_{i=1}^n (X_i^2 - (\bar{X})^2) \end{cases} \quad (4)$$

$$\Rightarrow \begin{cases} \alpha = \frac{n\bar{X}^2}{\sum_{i=1}^n (X_i^2 - (\bar{X})^2)} \\ \beta = \frac{n\bar{X}}{\sum_{i=1}^n (X_i^2 - (\bar{X})^2)} \end{cases} \quad (5)$$

2. (a) This is the “first case” on the “Important Confidence Intervals” list. Plugging in $\bar{x} = 14.3$, $s = 4.2$, and $n = 40$, we obtain a 95% confidence interval of **(13.00, 15.60)**.
- (b) No, we do not have enough information. When the sample size is small, you need to be able to assume that the sample is from a normal distribution in order to use the interval based on the t -distribution.
- (c) Having been told that the X_i are normal, and for $n = 10$, we are now in the “second case” on the “Important Confidence Intervals” list. Plugging in $\bar{x} = 14.3$, $s = 4.2$, and $n = 10$, we obtain a 95% confidence interval of **(11.30, 17.30)**.

Note that the Python command `sp.stats.t(df=9).ppf(0.975)`, where `sp` is the package `scipy`, can be used to obtain the appropriate quantile of the t distribution.

3. (a) If each $X_i \stackrel{IID}{\sim} \text{Poisson}(\lambda)$, then the likelihood function, the log-likelihood, and the derivative of the log-likelihood are, respectively:

$$\begin{aligned} L(\lambda) &= \prod_{i=1}^n e^{-\lambda} \frac{\lambda^{x_i}}{x_i!} = \frac{e^{-n\lambda} \lambda^{\sum x_i}}{\prod (x_i!)} \\ \ell(\lambda) &= -n\lambda + \left(\sum_{i=1}^n X_i \right) \log \lambda - \log \prod_{i=1}^n (x_i!) \\ \ell'(\lambda) &= -n + \frac{1}{\lambda} \sum_{i=1}^n X_i \end{aligned}$$

Setting this last expression equal to zero and solving for λ yields

$$\hat{\lambda}_{\text{MLE}} = \frac{1}{n} \sum_{i=1}^n X_i = \bar{X}.$$

Note that

$$\ell''(\lambda) = -\frac{1}{\lambda^2} \sum_{i=1}^n X_i$$

Since the X_i are all nonnegative, $\ell''(\lambda)$ is nonpositive everywhere, so $\hat{\lambda}_{\text{MLE}}$ is indeed the maximizer of the likelihood.

- (b) Yes, this is an unbiased estimator for λ since

$$E(\hat{\lambda}) = E(\bar{X}) = \frac{1}{n} \sum_i E(X_i) = \frac{1}{n} n\lambda = \lambda$$

- (c)

$$V[\hat{\lambda}_{\text{MLE}}] = V[\bar{X}] = \frac{1}{n^2} V\left[\sum_{i=1}^n X_i\right] = \frac{1}{n^2} nV[X_i] = \frac{\lambda}{n}$$

Hence, the standard error of $\hat{\lambda}_{\text{MLE}}$ is $\sqrt{\frac{\lambda}{n}}$. We would have to approximate this using $\sqrt{\frac{\hat{\lambda}}{n}}$. We could have also arrived at the same answer by using the Fisher Information. We would find that $I(\lambda) = 1/\lambda$.

4. See the code below. We find that approximately 56% of the simulated samples have the “adjusted” estimator closer to the true value than does the estimator based on the second moment. Although this is larger than 50%, it may not be as large as one might have expected.

```
import numpy as np

reps = 10000      # The number of repetitions
n = 20           # The sample size
lam = 10         # The value of lambda

# Define the two estimators

def adjust(x):
    return (1/np.mean(x))*(len(x)-1)/len(x)

def mom2est(x):
    return np.sqrt(2*len(x)/sum(x**2))

# Generate the random variates

rvs = np.random.exponential(1/lam, size=(reps,n))

# Calculate the estimates for each simulated sample

adjusts = np.apply_along_axis(adjust,1,rvs)
mom2ests = np.apply_along_axis(mom2est,1,rvs)

# For what proportion is the adjusted estimator better?

np.mean(abs(adjusts - lam) < abs(mom2ests - lam))
```

0.5606

5. (a)

$$E(\hat{\theta}_1) = E\left(\frac{4\bar{X}}{3}\right) = \frac{4}{3}E(\bar{X}) = \frac{4}{3}E(X) = (\text{given}) = \frac{4}{3} \cdot \frac{3}{4}\theta = \theta$$

(b)

$$\begin{aligned} E(\hat{\theta}_2) &= E(X_{(n)}) = \int_0^\theta x \frac{3n}{\theta^{3n}} x^{3n-1} dx = \frac{3n}{\theta^{3n}} \frac{x^{3n+1}}{3n+1} \Big|_0^\theta = \frac{3n}{3n+1} \theta \neq \theta \\ \Rightarrow \text{Bias}(\hat{\theta}_2) &= E(X_{(n)}) - \theta = \frac{3n}{3n+1} \theta - \theta = -\frac{1}{3n+1} \theta \end{aligned}$$

(c) Since $\hat{\theta}_1$ is unbiased,

$$\text{MSE}(\hat{\theta}_1) = V(\hat{\theta}_1) = V\left(\frac{4}{3}\bar{X}\right) = \frac{16}{9}V(\bar{X}) = \frac{16}{9n}V(X) = (V(X) \text{ is given}) = \frac{16}{9n} \frac{3}{80} \theta^2 = \frac{\theta^2}{15n}$$

$$\text{MSE}(\hat{\theta}_2) = V(\hat{\theta}_2) + [\text{Bias}(\hat{\theta}_2)]^2 = \frac{3n}{(3n+2)(3n+1)^2} \theta^2 + \frac{1}{(3n+1)^2} \theta^2 = \left[\frac{2}{(3n+2)(3n+1)} \right] \theta^2$$

$$\begin{aligned} \frac{\text{MSE}(\hat{\theta}_2)}{\text{MSE}(\hat{\theta}_1)} &= \frac{\text{MSE}(\hat{\theta}_2)}{V(\hat{\theta}_1)} = \frac{2 \cdot 15n}{(3n+2)(3n+1)} < 1, \text{ for } n > 2 \\ \Rightarrow \text{MSE}(\hat{\theta}_2) &< \text{MSE}(\hat{\theta}_1) \end{aligned}$$